# Predicting Linear Relational Decodability Without the Relation

**Anonymous ACL submission**

## Abstract

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations – near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We ask what determines that variation, and show it is substantially predictable from a readout that never sees the relation. We estimate a Jacobian lens from subject-to-subject differences under a single neutral prompt, at one fixed layer for every relation, using no relation template, no objects and no (subject, object) pairs and measure how highly it ranks the model's own answer. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.66$ ($p = 1.4 \times 10^{-4}$), rising to $0.75$ once object cardinality is partialled out. Neither cardinality nor the model's prior over the answer token accounts for the correlation, and the estimate is stable under leave-one-domain-out (0.57–0.70). Moving the unit of analysis from the relation to the subject — which holds prompt, answer space and cardinality fixed by construction — the lens predicts *which subjects within a relation* the LRE reproduces, at a median AUC of $0.69$ over 1,067 subjects in 13 relations, and survives a control for the model's own confidence. This suggests that linear decodability may be largely a property of what a subject's representation already carries, rather than of the relation being decoded.

## 1 Introduction

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Our diagnostic is a *Jacobian lens*: a readout that propagates a first-order approximation of the model's own computation from an intermediate layer to the output vocabulary. Lenses of this kind have recently been used to characterize what a model is internally disposed to say (Gurnee et al., 2026). Ours differs in what it is applied to: rather than a corpus-averaged Jacobian acting on raw activations, we evaluate the Jacobian at individual reference entities and apply it to *differences* between entity representations, under a single neutral prompt shared by every relation. The lens sees no relation template, no objects, and no (subject, object) pairs; the layer is fixed in advance rather than tuned per relation. We then ask how highly it ranks the model's own answer to the relation. That score correlates with LRE faithfulness across 28 relations at $\rho = 0.66$.

We make three contributions. (i) We introduce a relation-agnostic diagnostic of linear decodability and show it predicts LRE faithfulness. (ii) We show the prediction survives controls for object cardinality and answer-token frequency, holding out entire subject domains, and — moving to the subject as the unit of analysis — a control for the model's own confidence. (iii) We document a first-token scoring artifact that silently inflates faithfulness to a perfect score on relations with numeric objects, and propose a remedy.

## 2 Background

An LRE for relation $r$ approximates the model's computation as

$$o \approx \beta W s + b, \quad (1)$$

where $s$ is the hidden state at the subject's final token, $o$ the output state, and $W$, $b$ are estimated from a small set $\{s_i\}$ of reference subjects as the mean Jacobian and mean bias of the model's own map (Hernandez et al., 2024).

**1 <<< hs: define what $f$ is below? >>>**

The scalar $\beta$ corrects a systematic weakness of the first-order estimate. A Jacobian taken at a reference subject describes the model's map only in a neighborhood of that subject; extrapolated to a different subject it points in roughly the right direction but is too short. This matters because the prediction decomposes into a subject-dependent term and a constant: expanding $b$ gives

$$\beta W s + b \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i(s - s_i)\,], \quad (2)$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit $\beta$ per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$. Eq. 2 is also what makes the LRE affordable: each test subject costs $n$ Jacobian–vector products rather than a materialized $4096 \times 4096$ matrix.

*Faithfulness* is the rate at which decoding $\beta W s + b$ over the full vocabulary reproduces the *model's own* top-1 prediction – not the gold object. It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

## 3 A relation-agnostic Jacobian lens

For an entity $e$ (e.g., France) and a set of reference entities $\{i\}$ of the same type (e.g., Spain, Italy), we take the Jacobian–vector product **2 <<< hs: i find this confusing. can we simply cut it?: "of the model's map" >>>** at each reference and average:

$$v_e \;=\; \frac{1}{n} \sum_i J_i\,(s_e - s_i). \quad (3)$$

Here $J_i$ is the Jacobian of the model's map from the subject's hidden state at layer $\ell$ to the final-layer state at the last position, evaluated at reference $i$. We never form $J_i$ explicitly: a Jacobian–vector product $J_i u$ can be obtained for the cost of a constant number of forward and backward passes,[1] which is what makes Eq. 3 affordable at $d_{\text{model}} = 4096$. Each $J_i$ is evaluated under a *neutral* prompt, `"{} is known for"`, identical for every relation.

**Why differences.** The quantity we want is not what the model says about France, but what is *specific* to France among countries. A raw activation $s_e$ is dominated by what all entities of its type share – the type itself, the syntactic role, the prompt – and a lens applied to it returns mostly that shared content. Subtracting a same-type reference cancels it, leaving the entity-specific residual, and the Jacobian then carries that residual forward to the vocabulary. The lens therefore reads what this entity, as opposed to others of its kind, makes the model disposed to say.

We note that this differencing falls directly out of the rewriting of the Jacobian lens in Eq. 2, i.e., out of expressing the LRE as a constant plus a sum of $J_i(s - s_i)$ terms. Eq. 3 is that same quantity with the subject-independent anchor dropped and $\beta = 1$. This new way of looking at the Jacobian lens allows us to estimate the lens without any relation and ask whether it predicts an LRE that was fitted *with* one. This also distinguishes our lens from that of Gurnee et al. (2026), which averages a Jacobian over a corpus and applies it to raw activations.

We decode $v_e$ over the full vocabulary and record the rank of $m_e$, the model's own top-1 answer for relation $r$. Writing $\mathrm{rk}_{v_e}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\, \log_{10} \mathrm{rk}_{v_e}(m_e), \quad (4)$$

so higher $A_r$ means the lens agrees with the model more.

Three properties make $A_r$ non-trivial as a predictor. The prompt is neutral and shared, so no relation template enters. The differences in Eq. 3 are subject-minus-subject – *France* minus *Spain*, never *France* minus *Paris* – so no relation-specific

[1] We use the standard double-backward construction: differentiating $\langle u', f(h)\rangle$ with respect to $h$ with $u'$ held as a variable, then differentiating the result with respect to $u'$ in the direction $u$. This requires `float32` and eager attention, as fused attention kernels are not twice differentiable.

(subject, object) pair is used in estimation. The layer is not relation-specific either. It is fixed at 17 for all relations.[2]

Relation-specificity enters in exactly one place: the *target*. The model's top-1 is taken under the same 8-shot in-context prompt used to measure faithfulness, so the two quantities are commensurable.

## 4 Experimental setup

**Model.** Olmo-3-7B, in `float32` with eager attention, for the reason given in footnote 1.

**Relations.** The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter in advance using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"give is known for"` is a category error. No other relation was excluded.

**Measurement.** Both quantities are read off the same prompt, so that they are comparable. Following Hernandez et al. (2024), a relation is presented to the model *in context*: eight (subject, object) examples of that relation are concatenated, one per line, and the query subject is appended without its object, so that the model's next token is its answer. This is what defines "the model's own prediction" for both measures. The subjects of a relation are shuffled once and partitioned into three disjoint sets: eight in-context examples, eight reference subjects from which $W$ and $b$ are estimated, and the remainder as scored test subjects.

Faithfulness is then recomputed for all 28 relations under this protocol. Layer and $\beta$ are chosen on a $6 \times 6$ grid — six candidate layers for the subject state $(5, 11, 14, 17, 20, 23)$ crossed with six values of $\beta$ $(0.5, 1, 1.5, 2, 3, 5)$ — by the mean over the test subjects. Hernandez et al. (2024) likewise fit both per relation; the six layers span the network at roughly even spacing and the $\beta$ range brackets the values they report.

[2] Layer 17 is roughly half-way through the network, the region in which such lenses are usable at all (Gurnee et al., 2026), and is the modal best layer for the LRE itself in our sweep (10 of 28 relations; the rest spread over 5, 11, 14, 20 and 23). We stress that a single layer is used for *every* relation, so this choice cannot manufacture a correlation across relations: it can only make the diagnostic uniformly better or uniformly worse. Tuning the layer per relation would score higher and mean less.

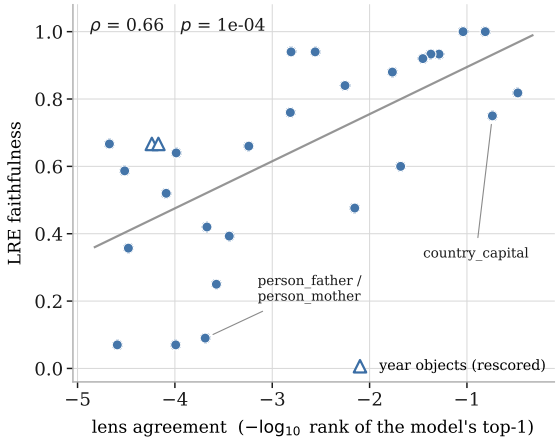



Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in §8.

The lens scores 11–20 subjects per relation, depending on how many remain once the in-context and reference sets are removed. For the relation-level results below, the lens and the LRE sample their splits independently, so the particular subjects entering each set differ between the two measurements. Both estimates are unbiased for the relation, but each carries sampling noise the other does not, which attenuates rather than inflates the correlation we report. The subject-level analysis of §6 requires the two to score the *same* subjects and uses a single shared split throughout.

Crucially, the in-context prompt is used only to obtain the *target*; the lens itself is always computed under the neutral prompt of §3.

## 5 Results

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.658$ ($p = 1.4 \times 10^{-4}$). With 28 points and a parametric $p$-value resting on assumptions we would rather not make, we also test the correlation by permutation: we shuffle the lens scores against the faithfulness values $10^4$ times, recompute $\rho$ each time, and count how often the shuffled $|\rho|$ matches or exceeds the observed one. It does so in $0.03\%$ of shuffles, so the association is not an artifact of the small sample.

**The correlation is not carried by cardinality.** Relations differ greatly in how many distinct objects they have; $k$, the mean number of subjects per object, ranges from 1.0 to 189. High $k$ makes faithfulness easier, and indeed $k$ predicts faithfulness on its own ($\rho = 0.43$, $p = 0.02$). But $k$ is essentially uncorrelated with the lens score ($\rho = -0.03$, $p = 0.87$; that is, no detectable relationship, which is what we want here, though with $n = 28$ we can only rule out a strong one). Partialling $k$ out *raises* the correlation to $\rho = 0.744$ ($p = 10^{-5}$): $k$ and the lens are independent predictors of faithfulness, so removing the variance attributable to $k$ sharpens rather than weakens the lens signal.

**Nor by answer frequency.** A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.03$) nor the lens score ($\rho = -0.02$), and partialling it out leaves the correlation unchanged at $0.658$; with both covariates (cardinality and answer frequency), $0.746$.

**How we partial.** All variables here are ordinal, so we work throughout on ranks. To partial out a set of covariates we rank-transform the lens score, faithfulness and each covariate; regress the ranked lens score on the ranked covariates by ordinary least squares with an intercept; do the same for ranked faithfulness; and report the Pearson correlation of the two residual vectors. This is the standard partial Spearman correlation, and it asks how much of the lens–faithfulness association survives once everything the covariates can linearly explain about either variable has been removed.

**Nor by one domain.** Eleven of the 28 relations have *person* subjects, raising the worry that one domain drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median $0.656$.

**Our findings do not rest on the flawed scoring.** Re-running under the corrected protocol of §8 — a single shared split and objects restricted to those their first token identifies — gives $\rho = 0.644$ ($p = 0.0005$) over the 25 relations that survive the restriction, against $0.658$ over 28 here.

**A within-domain check.** PERSON_FATHER and PERSON_MOTHER are separate relations over near-identical subject sets. They receive almost identical faithfulness ($0.09$, $0.07$) and almost identical lens scores ($-3.69$, $-3.99$), without the lens being told which relation it is looking at.

## 6 From relations to subjects

The evidence so far is 28 aggregate points, and a skeptic may reasonably suspect that both measures track some coarse property of a relation rather than anything about individual entities. Both quantities are, however, defined per subject: whether the LRE reproduces the model's answer for that subject, and where the lens ranks it. Asking whether the lens predicts *which subjects within a relation* the LRE gets right holds the relation, its prompt, its answer space and its cardinality fixed by construction — controlling by design for the covariates §5 partials out — and converts the claim from a statement about relations into one about entities, which is what the account asserts.

**Design.** The unit is a (relation, subject) pair. The outcome is whether the LRE's top-1 equals the model's top-1 for that subject; the predictor is $-\log_{10}$ of the rank at which the lens places that same token. A relation enters if it has at least 20 scored subjects, a faithfulness between $0.10$ and $0.90$, and both outcomes present — a relation at floor or ceiling has no within-relation variance to predict. Thirteen relations qualify, giving **1,067 subject-level rows**. Lens and LRE now take their split from one place and we verify by fingerprint that they scored identical subjects.

**Statistic.** Within each relation we compute the AUC: the probability that a subject the LRE reproduces has a higher lens score than one it misses. This is immune to between-relation variation by construction. We then test whether $\{\mathrm{AUC}_r\}$ lies above $0.5$ with a Wilcoxon signed-rank test across relations. A single pooled correlation over all 1,067 rows would be dominated by between-relation differences and would re-answer the question of §5.

**Result.** Median AUC is $\mathbf{0.688}$, above chance for 10 of 13 relations, with Wilcoxon $p = 0.011$ (Table 1). Permuting outcomes within each relation gives a null median AUC of $0.500$, matched in 0 of 2,000 permutations. Scoring all objects rather than only those whose first token identifies them (§8) gives a median of $0.686$ but $p = 0.084$: the significant version comes from correcting the scoring, not from dropping relations.

| relation | $n$ | hits | AUC | $p$ |
|---|---|---|---|---|
| superhero_person | 63 | 35 | 0.835 | 0.000 |
| landmark_in_country | 100 | 80 | 0.784 | 0.000 |
| person_mother | 100 | 11 | 0.766 | 0.004 |
| product_by_company | 100 | 68 | 0.715 | 0.001 |
| pokemon_evolutions | 20 | 5 | 0.707 | 0.197 |
| company_hq | 100 | 45 | 0.689 | 0.001 |
| company_ceo | 100 | 12 | 0.688 | 0.035 |
| person_plays_position_in_sport | 100 | 84 | 0.686 | 0.019 |
| person_plays_instrument | 100 | 73 | 0.627 | 0.053 |
| star_constellation | 100 | 35 | 0.617 | 0.055 |
| person_occupation | 100 | 66 | 0.494 | 0.922 |
| superhero_archnemesis | 55 | 18 | 0.467 | 0.700 |
| person_university | 29 | 10 | 0.268 | 0.046 |
| median | 1067 | | **0.688** | 0.011 |

Table 1: Within-relation discrimination. For each relation, AUC is the probability that a subject the LRE reproduces has a higher lens score than one it misses; chance is $0.5$. The final row gives the median AUC and the Wilcoxon signed-rank test of $\{\mathrm{AUC}_r\}$ against $0.5$.

| model | lens $\beta$ | confidence $\beta$ |
|---|---|---|
| lens only | $+1.90$ | — |
| lens + top-1 probability | $+1.33$ | $+2.07$ |
| lens + entropy | $+1.42$ | $-1.56$ |
| confidence only | — | $+2.44$ |

Table 2: The lens effect with and without a control for model confidence. All coefficients have $p < 10^{-4}$.

**Is it just confidence?** If the model is peaked on a subject, both the LRE and the lens may find it easier, and the association would be real but uninformative. We measure confidence as the model's top-1 probability at the target position, and fit a fixed-effects logistic regression stratified by relation, so that only within-relation variation contributes. Predictors are within-relation percentile ranks, so a coefficient is the log-odds change across a relation's own range.

The lens effect survives: an odds ratio of $6.7\times$ across the within-relation range alone, $3.8\times$ with confidence partialled out — attenuated by 30%, still at $p < 10^{-4}$. Entropy agrees, with the expected sign reversal.

But confidence is the *stronger* predictor, at $11.5\times$ alone, and the two are correlated within relation ($\rho = 0.32$). The honest statement is therefore that the lens predicts which subjects the LRE reproduces independently of the model's confidence, that confidence predicts it better, and that roughly a third of the raw lens effect was confidence. The asymmetry that makes the lens interesting anyway is that it is computed without the relation, whereas confidence is measured under the relation's own prompt.

**How large is the effect?** Most of the variation in LRE success lies *within* relations, not between them: an analysis of variance on the binary outcome puts $74.7\%$ within and $25.3\%$ between, and the latent-scale intraclass correlation from a random intercept per relation agrees at $29.3\%$ between. The subject-level analysis is therefore addressing the larger component rather than a residual.

It explains a modest share of it. Taking per-relation intercepts as the baseline, McFadden's pseudo-$R^2$ within relations is $0.047$ for the lens, $0.075$ for confidence and $0.095$ for the two together. The effect is real and robust but small, and the bulk of within-relation variation in LRE success is explained by neither.

**The scoring penalizes the lens.** First-token rank scoring requires the lens to surface the exact answer token, and penalizes it for producing a synonym or a near neighbor. On a control set of 80 famous people with a single unambiguous occupation each, the lens places COMPOSER at median rank 3 for composers, but WRITER at rank 244 for writers — whose readings are *novels*, *prose*, *poetry*, *literary*, all correct and none of them the scored token. The effects we report are therefore lower bounds by an amount we can demonstrate but not yet quantify.

## 7 Where the account degrades

What the lens reads is what an entity is *salient* for. On the account proposed here that is not a confound to be excused but the mechanism itself: a relation is linearly decodable to the extent that its answer is already part of what the subject's representation makes the model disposed to say. The relations that deviate from the trend are therefore exceptions to the account, not failures of measurement, and it is worth asking whether they are unstructured.

**Salience, measured independently of the lens.** The account above would be circular if salience were read off the lens, since the lens is itself a salience readout. We therefore measure it by the model's own free generation. For each test subject we greedily generate 90 tokens from each of `"{} is best known for"` and `"Here are some facts about {}. First,"`, and record whether any distinctive word of the gold object appears in either continuation.[3] This is a forward pass under

[3] Titles, particles and words shared with the subject's own name are excluded, so that *The* Joker is not matched by any

different prompts, and involves no Jacobian.

Call this rate the object's *prominence*. Across the 14 relations for which it can be measured it ranges from 1.00 (CITY_IN_COUNTRY) and 0.50 (COUNTRY_CAPITAL_CITY) down to 0.13 (PERSON_FATHER), 0.05 (SUPERHERO_ARCHNEMESIS) and 0.02 (PERSON_MOTHER). It predicts how often the lens surfaces the gold object in its top five: Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). The salience interpretation is thus supported by a measurement that is independent of the quantity it explains.

One relation disobeys: STAR_CONSTELLATION has prominence 0.60 yet the lens surfaces the constellation for only 5% of its stars. We suspected an artifact, since a Bayer designation contains the constellation in the Latin genitive (*Eta Persei* for *Perseus*), which our word filter does not remove; but prominence on the 19 subjects without such overlap is 0.63, slightly higher rather than lower, so this explanation fails. What the lens returns for these subjects is category vocabulary — *constellation*, *brightness*, *stellar* — rather than the particular constellation, which suggests the same first-token scoring problem discussed in Section 8. Excluding the relation would raise the correlation to 0.725; we report the value with it included.

They are not. Ordering relations by their residual from the fitted trend, both tails are organized by the size of the answer space. The largest positive residuals – faithful despite a low lens score – are LANDMARK_ON_CONTINENT ($+0.30$, $k = 189$), NAME_GENDER ($+0.29$, $k = 9.5$) and PERSON_NATIVE_LANGUAGE ($+0.26$, $k = 30.6$). In each, the LRE chooses among a handful of possible answers, and can be faithful without the answer being salient for any particular subject: no first name is chiefly "known for" its gender, but there are only two answers to pick between. The largest negative residuals – unfaithful despite the lens – are PERSON_FATHER ($-0.43$), PERSON_MOTHER ($-0.41$) and COMPANY_CEO ($-0.32$), all with $k \approx 1$ and answer spaces in the hundreds, where faithfulness is near its floor whatever the representation carries.

The account thus holds in the middle of the cardinality range and degrades predictably at both ends, in opposite directions. This is the same variable whose removal raises the correlation to 0.744, and it gives that number an interpretation: partialling out $k$ removes the principal source of deviation from the salience account rather than merely a nuisance correlate.

## 8 A first-token scoring artifact

Both relations with numeric objects (PRESIDENTS_BIRTH_YEAR, PRESIDENTS_ELECTION_YEAR) initially scored faithfulness 1.000. This was spurious. In this tokenizer `" 1743"` segments as `[' ', '174', '3']`: digits never merge with a preceding space. First-token scoring therefore compared the model's top-1 against a bare space token on both sides, and faithfulness 1.000 meant only that the LRE and the model both emit a space.

The fix is to place the space in the prompt, so that the query ends with it and the next token is content. Rescored, the two relations give faithfulness 0.67 and lens scores of $-4.24$ and $-4.17$; the model does predict the year correctly once the space is consumed. Including the uncorrected values would have depressed the reported correlation from 0.658 to 0.427.

**The problem is more general.** Numeric objects are the extreme case, but any two objects sharing a first token are indistinguishable to a metric that scores only that token: *University of Chicago* and *University of Michigan* both reduce to UNIVERSITY. Measuring the collapse rate — the fraction of a relation's distinct objects that share a first token with another — **6 of 40 relations exceed 25%**, led by PERSON_MOTHER (61%), PERSON_FATHER (58%) and PERSON_UNIVERSITY (45%, where 29 of 69 objects begin with UNIVERSITY).

Restricting each relation to objects whose first token identifies them repairs this on the gold side, and is the scoring used in §6. It is only a partial remedy, because faithfulness never uses gold: it compares the LRE against the *model's* top-1, so what inflates it is the model's own answer being an easy token. In PERSON_UNIVERSITY, after filtering, the model still answers UNIVERSITY for 7 of 29 subjects and the LRE agrees on every one. A complete fix would condition on the prediction being distinctive.

The direction of the bias differs by relation. Where the collapse is on the gold side it makes

occurrence of "the" and *Johann Georg Kant* is not matched by a passage about Immanuel Kant. The lens's own prompt cannot be used here: `"{} is known for"` makes this model emit a multiple-choice quiz rather than prose, so the object is never named.

faithfulness easier — 41 distinct fathers all counting as JOHN — so our PERSON_FATHER figure is an upper bound, and the corrected value is lower still ($0.07 \rightarrow 0.05$). Where the model itself emits the shared token, the collapse manufactures agreement that is not retrieval at all.

We flag all of this because the failure is silent: it produces a perfect or inflated score rather than an obvious error, and it will affect any first-token evaluation whose answers share prefixes — numeric, titled (*Doctor Doom*, *The Joker*) or generic (*University of X*).

## 9 Limitations

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits.

The subject-level analysis of §6 carries several caveats of its own. Model confidence predicts LRE success better than the lens does ($11.5\times$ against $6.7\times$), and explains more of the within-relation variance ($0.075$ against $0.047$). The lens contributes independently of it, but neither explains much: together they account for under a tenth of the within-relation variation in LRE success. Faithfulness there is estimated from a single split per relation, and a single split is noisy — one relation scored $0.39$ and $0.21$ on two draws at identical test-set size — so the outcome variable is measured with error that attenuates the association. The thirteen qualifying relations all sit in the middle of the cardinality range, where the account is expected to work, making this a fair test of the mechanism but not of its boundaries. And the inclusion criteria screen on faithfulness and test-set size, neither of which catches a relation like PERSON_OCCUPATION where two answers cover 65% of subjects; screening on the concentration of the model's answer distribution would be better, and would have to be fixed in advance.

The account has a scope condition we can state but not yet explain (§7). It is weakest where the answer space is very small, since an LRE can then be faithful by elimination rather than by reading anything from the subject, and where the answer space is very large and each answer near-unique, since faithfulness is then close to its floor regardless. We show that these are the two tails of a single variable, but we do not have a principled account of where the boundaries lie, and with 28 relations we cannot fit one.

We also probe salience through a single neutral prompt, `"{} is known for"`. That phrasing asks what an entity is *noted* for, which need not coincide with everything its representation carries; attributes such as gender may be well encoded without being something a name is "known for". A different neutral prompt, or an average over several, might track salience more broadly.

Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these. Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

## 10 Conclusion

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful.

The reading we favor is that salience is the mechanism rather than a confound: a relation admits a faithful linear operator to the extent that its answer is already part of what the subject's representation makes the model disposed to say. On that view the wide spread of LRE faithfulness across relations is less a fact about the relations than about the entities they are asked about – and the exceptions are informative, falling at the two ends of the answer-space range where faithfulness can be achieved, or prevented, without the subject's representation mattering much either way.

## References

Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, T. Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. 2026. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*.

Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2024. Linearity of relation

decoding in transformer language models. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

## A All relations

Table 3 gives faithfulness and the lens score for every relation in the study.

| relation | $k$ | faith. | $A_r$ | relation | $k$ | faith. | $A_r$ |
|---|---|---|---|---|---|---|---|
| country_language* | 1.7 | 1.00 | $-1.04$ | presidents_election_year† | 1.1 | 0.67 | $-4.17$ |
| country_largest_city* | 1.0 | 1.00 | $-0.81$ | product_by_company | 17.4 | 0.66 | $-3.24$ |
| landmark_on_continent | 189.4 | 0.94 | $-2.81$ | person_plays_position_in_sport | 68.0 | 0.64 | $-3.99$ |
| person_native_language | 30.6 | 0.94 | $-2.56$ | person_band_lead_singer* | 1.0 | 0.60 | $-1.68$ |
| name_birthplace | 3.9 | 0.93 | $-1.29$ | person_university | 1.3 | 0.59 | $-4.52$ |
| name_religion | 6.2 | 0.93 | $-1.37$ | company_hq | 4.1 | 0.52 | $-4.09$ |
| person_plays_pro_sport | 63.6 | 0.92 | $-1.45$ | superhero_person | 1.0 | 0.48 | $-2.15$ |
| person_plays_instrument | 85.5 | 0.88 | $-1.77$ | star_constellation | 11.7 | 0.42 | $-3.67$ |
| landmark_in_country | 9.2 | 0.84 | $-2.25$ | pokemon_evolutions | 1.0 | 0.39 | $-3.44$ |
| city_in_country | 1.3 | 0.82 | $-0.48$ | country_currency | 1.3 | 0.36 | $-4.48$ |
| person_occupation | 26.5 | 0.76 | $-2.81$ | superhero_archnemesis | 1.1 | 0.25 | $-3.57$ |
| country_capital_city* | 1.0 | 0.75 | $-0.74$ | person_father | 1.0 | 0.09 | $-3.69$ |
| name_gender* | 9.5 | 0.67 | $-4.67$ | company_ceo | 1.0 | 0.07 | $-4.59$ |
| presidents_birth_year† | 1.3 | 0.67 | $-4.24$ | person_mother | 1.0 | 0.07 | $-3.99$ |

Table 3: All 28 relations, sorted by LRE faithfulness. $k$ is the mean number of subjects per distinct object; $A_r$ is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (§8); *fewer than ten test items, so faithfulness is coarsely quantized.